

1. Simulate 30 rolls with =RANDBETWEEN(1,6). What is the probability of rolling a 3 exactly 5 times? (Hint: Use BINOM.DIST)

Solution:

This problem is based on the **Binomial Distribution** because:

1. There are a fixed number of rolls (30).
2. Each roll is independent.
3. Each roll has only two outcomes — “Success” (getting a 3) or “Failure” (not getting a 3).
4. The probability of success stays constant at $p = \frac{1}{6}$.

Formula:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Given:

- $n = 30$
- $k = 5$
- $p = \frac{1}{6}$

Step-by-step:

$$P(X = 5) = \binom{30}{5} \left(\frac{1}{6}\right)^5 \left(\frac{5}{6}\right)^{25}$$

In Excel:

=BINOM.DIST(5, 30, 1/6, FALSE)

Final Answer:

$$P(X = 5) = 0.1921 \text{ or } 19.21\%$$

So, the probability of rolling a 3 exactly 5 times in 30 rolls is **19.21%**.

2. Generate 100 values in Excel using the continuous uniform distribution RAND() and plot a histogram. Describe the shape of the distribution.

Solution:

The RAND() function in Excel generates numbers between **0 and 1**, where each value in this range is equally likely.

This follows a **Continuous Uniform Distribution**.

In Excel:=RAND().

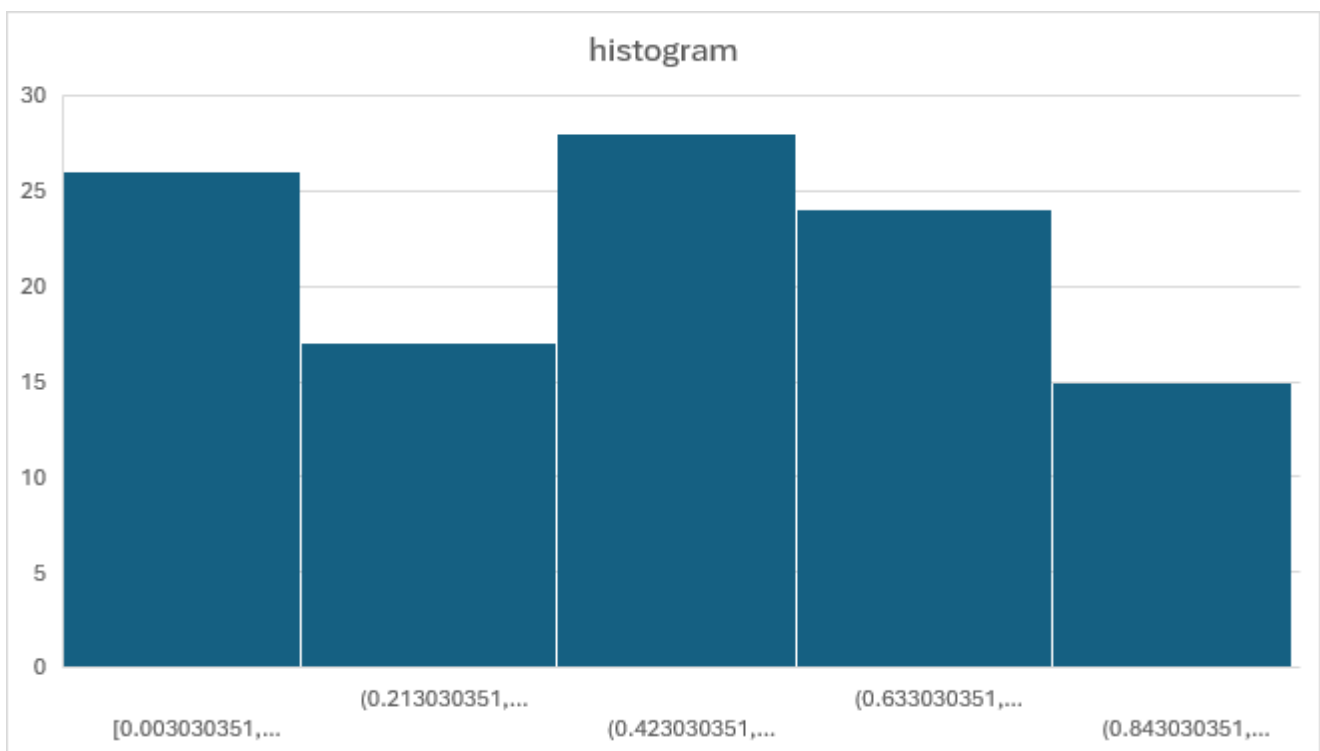
Explanation:

A uniform distribution means all values occur with equal probability.

So, when you plot the histogram, it should look **flat** — meaning no particular number range appears more often than another.

Final Answer:

The histogram is **roughly flat or rectangular**, showing that all values between 0 and 1 occur with almost equal frequency.



3. A dataset has a mean of 50 and a standard deviation of 5. What percentage of values lie between 45 and 55 if the data follows a normal distribution?

Solution:

This question uses the **Empirical Rule** (68–95–99.7 Rule) for Normal Distribution.

Empirical Rule:

- 68% of data lies within ± 1 standard deviation.
- 95% lies within ± 2 standard deviations.
- 99.7% lies within ± 3 standard deviations.

Given:

- Mean (μ) = 50
- Standard deviation (σ) = 5

Range:

$$45 = 50 - 5 = \mu - 1\sigma$$

$$55 = 50 + 5 = \mu + 1\sigma$$

So, the range from 45 to 55 covers **one standard deviation** from the mean.

Final Answer:

Approximately **68%** of the values lie between 45 and 55.

4. What is the concept of standardization (z-score), and why is it important in data analysis? Explain the formula and how standardization transforms a dataset.

Solution:

Concept:

Standardization (or z-score) converts raw data into a common scale that shows how far each value is from the mean, measured in standard deviations.

Formula:

$$z = \frac{x - \mu}{\sigma}$$

Where:

z = standardized value

x = raw score

μ = mean

σ = standard deviation

Why It's Important:

1. It allows comparison of values from different datasets or scales.
2. It helps identify **outliers** (values far from the mean).
3. It's essential for many statistical and machine learning methods.

Transformation Effect:

After standardizing:

- The new **mean becomes 0**.
- The new **standard deviation becomes 1**.

Final Answer:

Standardization transforms data into z-scores, making them comparable and centered around zero.

5. What is Kurtosis and its types?

Solution:

Concept:

Kurtosis measures how heavy or light the tails of a distribution are compared to a normal distribution — in simple terms, it shows **how peaked or flat** a distribution is.

Types of Kurtosis:

1. Mesokurtic ($K = 3$)

- Same as a normal curve.
- Moderate tails and peak.

2. Leptokurtic ($K > 3$)

- Sharper peak and heavier tails.
- Indicates more outliers.

3. Platykurtic ($K < 3$)

- Flatter peak and lighter tails.
- Indicates fewer outliers.

Final Answer:

Kurtosis describes the "tailedness" of a distribution.

Types are **Mesokurtic**, **Leptokurtic**, and **Platykurtic**.

6. Explain why the uniform distribution is a good model for the outcome of rolling a fair die.

Solution:

A **uniform distribution** means every possible result has an equal chance of happening.

For a fair die:

- Outcomes: 1, 2, 3, 4, 5, 6
- Each outcome has the same probability:

$$P(X = i) = \frac{1}{6}, i = 1, 2, \dots, 6$$

Since all outcomes are equally likely, this matches the definition of a **discrete uniform distribution**.

Final Answer:

A fair die follows a **discrete uniform distribution** because each of its six faces has an equal probability of appearing.

7. Use Excel to compute the probability of getting at least 8 successes in 15 trials with success probability 0.5.

Solution:

We are finding:

$$P(X \geq 8) = 1 - P(X \leq 7)$$

where $X \sim \text{Binomial}(n = 15, p = 0.5)$.

In Excel:

=1 - BINOM.DIST(7, 15, 0.5, TRUE)

Explanation:

The cumulative probability up to 7 gives $P(X \leq 7)$.

Subtracting this from 1 gives the probability of getting **8 or more** successes.

Final Answer:

$$P(X \geq 8) = 0.5$$

This makes sense because with $p = 0.5$, the distribution is symmetrical — half the time you'll get 7 or fewer successes, and half the time 8 or more.

8. How does log transformation help in stabilizing variance and making data more normally distributed?

Solution:

A **log transformation** is used when data is highly skewed (especially right-skewed).

1. Stabilizing Variance:

Large values get compressed more than small ones.

For example:

$$\log(100) = 2, \log(1000) = 3, \log(1,000,000) = 6$$

This compression reduces the spread of data, making variance more stable.

2. Making Data Normal:

Right-skewed data has a long tail on the higher side.

Taking logs “pulls in” the tail, making the data more **symmetric** and closer to a **bell-shaped curve**.

Final Answer:

Log transformation:

1. **Stabilizes variance** by compressing large values.
2. **Makes data more normally distributed** by reducing right-skewness.